LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Practice Pack: Tissues

Chapter / Topic: Topic-based: use with whichever Class 9 book your students follow

For teacher use only. Do not distribute to students.

LEVEL 1: CONCEPTS

Q1. What is a tissue?

Answer: A group of cells that are similar in structure and work together to perform a particular function.

Q2. Name the three types of meristematic tissue, according to where they are found.

Answer: Apical (at root and stem tips), lateral (increases girth) and intercalary (at the base of leaves or internodes).

Q3. Which plant tissue gives flexibility to leaves and stems?

Answer: Collenchyma.

Q4. Name the fluid matrix of blood and the tissue that connects muscle to bone.

Answer: Plasma; tendon.

Q5. Which type of muscle is found in the heart? Is it voluntary or involuntary?

Answer: Cardiac muscle. It is involuntary.

LEVEL 2: BOARD-STYLE QUESTIONS

Q6. Differentiate between parenchyma, collenchyma and sclerenchyma (any two features each).

Answer: Parenchyma: living, thin-walled, loosely packed cells. Stores food; with chlorophyll it is chlorenchyma, and with air spaces in aquatic plants it is aerenchyma. Collenchyma: living, elongated cells, thickened at the corners. Gives flexibility and support (leaf stalks). Sclerenchyma: dead cells with walls thickened by lignin. Makes plant parts hard and stiff (husk of coconut, fibres).

Q7. Name the elements of xylem and phloem.

Answer: Xylem: tracheids, vessels, xylem parenchyma and xylem fibres. Phloem: sieve tubes, companion cells, phloem fibres and phloem parenchyma.

Q8. Give three differences between striated, smooth and cardiac muscles.

Answer: Striated: voluntary; long, cylindrical, unbranched and multinucleate cells; found in limbs (attached to bones). Smooth: involuntary; spindle-shaped, uninucleate, no striations; found in the alimentary canal, blood vessels and iris. Cardiac: involuntary; cylindrical, branched and uninucleate; found only in the heart, and contracts rhythmically all through life.

Q9. What is the difference between a ligament and a tendon?

Answer: A ligament connects bone to bone; it is very elastic and strong. A tendon connects muscle to bone; it is fibrous, very strong, and has limited flexibility.

Q10. What are the functions of epithelial tissue? Give two examples of where it is found.

Answer: It forms a protective covering over the body and lines organs and cavities, and it controls the exchange of materials. Examples: the skin, the lining of the mouth, the alveoli of the lungs, kidney tubules.

LEVEL 3: CHALLENGE (FOUNDATION)

Q11. Why does a tree trunk grow thicker every year, while a grass stem does not? Name the tissue responsible.

Answer: A tree has lateral meristem (cambium), which keeps dividing and increases the girth of the stem. Grasses have no lateral meristem; they grow mainly by intercalary meristem, which increases length, not thickness.

Q12. Neurons can be over a metre long. Why does their shape suit their function?

Answer: Neurons carry messages (nerve impulses) over long distances in the body. A long axon carries the impulse far, and branched dendrites receive signals from many other cells, giving fast communication between distant parts of the body.

Q13. Water hyacinth floats on water. Which tissue helps it, and how?

Answer: Aerenchyma: parenchyma with large air cavities. The air inside gives buoyancy, so the plant floats.